KUKA Industrial Automation

Mechatronics Implementation Engineer | Pepper Equipment Corporation | Monte Vista, Colorado
September 19, 2022 - March 1, 2023

Project status: Agricultural AI-robotics research and development. The company discontinued the project before prototype completion because projected time and resource requirements exceeded the stakeholders' willingness to continue. This description covers work performed during development.

1) Project overview

The project investigated AI-assisted robotic handling for an agricultural potato-grading application. A KUKA Delta robot operated above a product conveyor, with a pneumatic gripper and external operating controls. The engineering work required connecting robot software, AI-system interface code, conveyor behavior, electrical I/O, and pneumatic/mechanical hardware.



Figure 1. Robot and potato conveyor during development. Author-supplied workplace photo, November 2022.

Individual responsibilities

The company work-experience statement confirms programming and testing of interface code between the AI system and KUKA Delta robot, plus installation and testing of pneumatic and mechanical systems associated with the robot cell. The résumé adds robot/conveyor programming, external-mode controls, and switch wiring. Additional company fabrication assignments are described separately in Section 6.

2) Application and robot configuration

The physical arrangement placed a Delta robot above a roller conveyor carrying potatoes. The robot program coordinated picking and placement through named motion targets, gripper commands, conveyor-related state, and chute positions. Photographs support the system layout but do not specify a complete as-built equipment list or control-cabinet design.

Subsystem	Role and evidence
KUKA Delta robot	Motion and tool manipulation; visible in workplace photos.
Conveyor	Product presentation and motion-related program logic; roller transport visible.
Pneumatic end effector	Pickup/release actuation; valve, tubing, and gripper functions visible.
Robot controller/pendant	Program editing, I/O inspection, and test operation.
External operator controls	Four-switch interface plus emergency-stop and operator-safety switches, author-reported.
AI-system interface	Interface code programming/testing confirmed by the company statement; upstream model details not supplied.



Figure 2. Robot and conveyor work area with pneumatic/electrical routing. This is a development-stage photo, not a completed-system acceptance record.

The ZIP supplies a KR 3 D1200 HM datasheet: four controlled axes, ceiling mounting, 1200 mm working diameter, 3 kg rated payload, and manufacturer pose repeatability of +/-0.1 mm. These are model specifications, not measured project results. The installed robot/controller identity is not independently established from that sheet alone.

3) KRL interface and conveyor programming

The author received KUKA Robotics Language training and KRL certification from KUKA Robotics Corporation, according to the company statement. The current résumé separately lists Robot Programming 1 and Robot Programming 2 at KUKA College. This training supported programming, integration, and testing responsibilities within the agricultural robotics project.

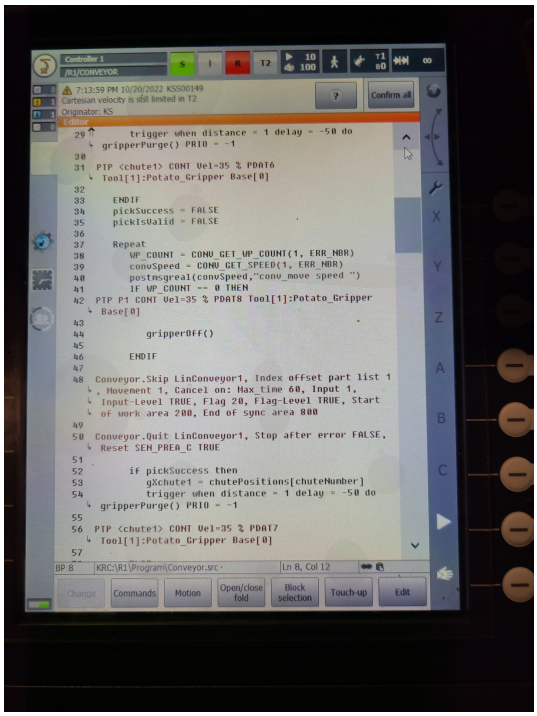


Figure 3. Pendant photograph of Conveyor.src. Visible logic includes conveyor queries, pick-state handling, motion statements, and gripper functions.

What the program evidence shows

The photographed excerpt queries conveyor workpiece count and speed, resets pickSuccess/pickIsValid state, uses conditional handling when no workpieces are available, and chooses chutePositions[chuteNumber] after a successful pick. PTP statements with CONT, tool references, and gripperOff/gripperPurge calls connect motion and tool actions.

The company statement establishes responsibility for AI-to-robot interface code. The photographs alone do not reveal the complete upstream AI architecture, message protocol, classification model, or end-to-end transaction format. Those implementation details are not invented here. The available evidence is program excerpts rather than a complete exported KRL project.

4) Motion sequencing and external control

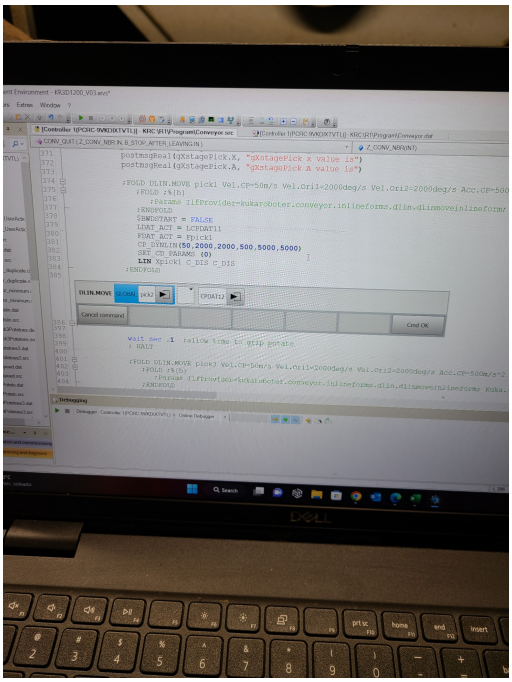


Figure 4. Development-environment photograph of Conveyor.src. The view includes DLIN.MOVE inline forms, LIN motion, motion-data assignments, and a wait statement annotated for gripping.

Motion and process coordination

Robot and conveyor programming aimed to provide smooth, coordinated handling. The code photo shows application-level motion data and timing instructions, but a displayed velocity setting is not a measured cycle rate. No before/after throughput, positioning error, or production acceptance figure is supplied.

Automatic external operation

The author wired, configured, and programmed an external operating interface using four switches, an emergency-stop switch, and an operator-safety switch. This work linked field wiring to robot-controller configuration and intended operating sequences. Exact functions for all four switches are not specified in the supplied description.

The image collection contains I/O screens, connector/switch wiring, and configuration views. They document hands-on implementation rather than a complete verified circuit. Safety certification, circuit redundancy, risk-assessment results, and a validated performance level are not established by the uploaded materials.

5) Pneumatic and mechanical integration

The company statement confirms installation and testing of both pneumatic and mechanical systems associated with the robot cell. Electrical outputs and pneumatic components had to support the programmed gripper actions, while mechanical integration provided the physical operating arrangement. The supplied evidence supports a multidisciplinary implementation role rather than software-only work.

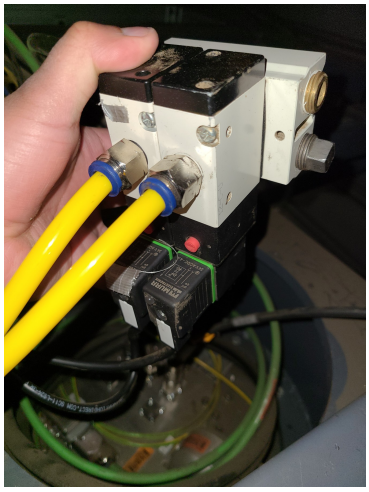


Figure 5. Pneumatic valve/solenoid assembly, tubing, and connected wiring. Component settings are not inferred from the image.



Figure 6. Digital pneumatic instrument in the workcell. The photograph documents hardware presence; it is not a calibration certificate or recorded process-pressure test.

The functional chain can be described as robot program -> controller I/O -> pneumatic actuation -> gripper response. Conveyor state and robot motion also influence the handling sequence. This is a role-level explanation, not an electrical schematic. Complete signal mappings and timing measurements would be needed for reproduction or commissioning documentation.

6) Additional duties and project outcome

Workcell mechanical work versus other fabrication

Robot-cell mechanical installation and testing belong to the AI-robotics project. In contrast, manufacturing mechanical parts through MIG welding, grinding, drilling, and other shop equipment was additional work performed for the company on other assignments. Those fabrication duties are not presented as part of this robot-workcell procedure.



Figure 7. Integration-stage wiring and pneumatic routing in the cell. No dedicated fabrication photograph was identified in the submitted project materials.

Outcome and professional contribution

The project closed during R&D; before a completed prototype was delivered. Its closure does not erase the performed engineering work: the company confirms interface-code programming/testing and pneumatic/mechanical installation/testing. It also states that performance met management’s satisfaction and recognizes initiative, cooperation, effort, and commitment.

The document uses the employment end date March 1, 2023, as explicitly corrected by the author. The pasted company paragraph gives February 28, 2023; the author-requested correction takes precedence in this description. The closure context is retained without presenting an unfinished project as a completed production deployment.

7) Engineering lessons and supporting evidence

Capabilities demonstrated

The assignment combined KRL application work, AI/robot interface testing, conveyor coordination, controller I/O, wiring, pneumatic actuation, and mechanical integration. The central engineering value is connecting software decisions to a physical handling process and investigating how the interfaces behave during development.

Documentation needed for a future continuation

A successor team would need the controller backup, versioned KRL source, upstream AI interface specification, I/O mapping, mechanical drawings, pneumatic schematic, and test records. Useful acceptance measures would include cycle time, successful picks per product condition, communication failures, controlled-stop behavior, and recovery after a missed pick.

Current evidence limits

The archive contains 38 workplace photographs and one robot datasheet, not a full report or runnable source export. Photos show selected code/configuration and hardware states but cannot establish every test step. No numerical production improvement is asserted. Manufacturer specifications remain distinct from application-level performance.

Source register

KUKA Industrial Automation.zip: Images/20221006_134812.jpg through IMG-20230210-WA0001.jpg and KUKA Delta Robot Data Sheet.pdf (V6.1, 27 May 2022). Additional user-selected program photo: 20221208_094402.jpg. Résumé: Dak Amarasinghe Resume(20261030-094031).pdf. Pasted company work-experience statement and author clarifications establish responsibilities, training, project closure, separate fabrication duties, and the requested end date.

Conclusion

This project demonstrates applied industrial mechatronics experience within an agricultural AI-robotics development effort. The strongest substantiated contributions are AI-to-KUKA interface programming/testing and integrating pneumatic, mechanical, and operator-control elements around the robot cell.